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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Pentagrammers
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	Orbyte Smart Reorder Assistant
Name (Leader)	CHA ZHI YI
Email & Phone (Leader)	chazyee0306@gmail.com 0169236287
University Affiliation(s)	Universiti Teknikal Malaysia Melaka (UTeM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small neighbourhood grocery shops, known as *kedai runcit* in Malaysia, play an important role in providing daily necessities to local communities. However, many small grocery shop owners have limited staff and time to manage their inventory efficiently.

For example, Pak Abu runs a small *kedai runcit*. After closing his shop at 10 PM, he spends time checking shelves, recording low-stock products, and contacting different suppliers through WhatsApp. This repetitive process takes up valuable time and may lead to human errors, forgotten orders, and delayed restocking. If essential products such as cooking oil, bottled water, or instant noodles run out of stock, the shop may lose sales and disappoint customers.

These challenges are particularly important for small businesses with limited resources and no dedicated inventory management staff. Unlike larger retailers, they may not be able to afford complex or expensive inventory systems.

A simple and reliable automation solution could help *kedai runcit* owners reduce repetitive administrative work, maintain more accurate stock records, and contact suppliers in a timely manner. By improving inventory management and reducing avoidable stock shortages, the solution aims to help small grocery businesses use their time and resources more productively.

This problem aligns with Sustainable Development Goal 8 (SDG 8), particularly Target 8.2 on productivity through technological upgrading and innovation, and Target 8.3 on supporting entrepreneurship and the growth of micro, small, and medium-sized enterprises.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The Orbyte Smart Reorder Assistant is an AI-assisted inventory automation system designed for small grocery shops (kedai runcit) in Malaysia. It aims to reduce the repetitive work of monitoring stock levels and preparing supplier messages.

When a sale is recorded through barcode scanning or manual input, the system updates the product quantity in the MySQL database. The JavaScript backend, running on Node.js, checks whether the remaining stock has fallen below the minimum level set by the shop owner.

When low stock is detected, the system retrieves the relevant product and supplier information. The Gemini API then generates a contextual WhatsApp replenishment message, including the product name, current stock level, and suggested reorder quantity. Rule-based logic determines when a low-stock alert should be triggered, while AI helps prepare clear and relevant supplier communication.

Human control is central to the workflow. The shop owner can review, edit, approve, or reject the generated message through the dashboard. Only an approved message can proceed to the configured sending process. The system records the action and its outcome in an activity log to support transparency and accountability.

The proposed technology stack includes JavaScript for the frontend, Node.js for backend processing and API endpoints, MySQL for inventory and supplier data, the Gemini API for AI-generated messages, and Replit for hosting and deployment.

By combining rule-based automation with AI-assisted communication, the solution aims to reduce repetitive tasks, improve stock-management consistency, and support SME productivity. It aligns with SDG 8, particularly Targets 8.2 and 8.3, by helping small businesses use technology to improve operations and support business growth.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

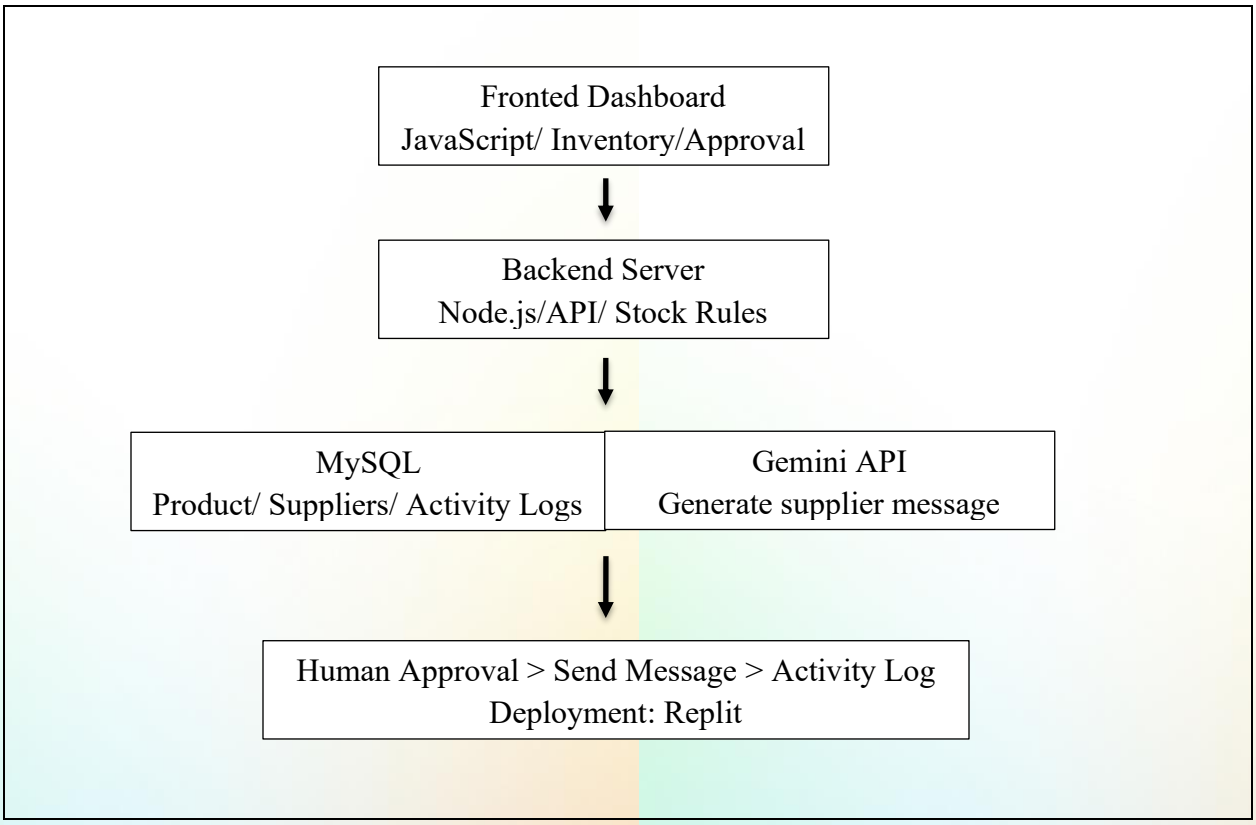
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies and Tools: The Orbyte Smart Reorder Assistant will use the following technologies to build a functional prototype:

- **Frontend:** JavaScript for the inventory dashboard, low-stock notifications, and human approval interface.
- **Backend:** Node.js to handle API requests, update inventory, check stock thresholds, and coordinate the replenishment workflow.
- **Database:** MySQL to store product details, barcode identifiers, supplier information, and activity logs.
- **Generative AI:** Google Gemini API to generate contextual WhatsApp replenishment messages based on product and supplier information.
- **Cloud Platform:** Replit to host and deploy the Node.js application and provide access to the working prototype.
- **Supplier Communication:** WhatsApp message generation and approval workflow. Actual message delivery will depend on the messaging integration implemented.

System Architecture: The frontend sends inventory updates to the Node.js backend through API endpoints. The backend retrieves and updates product records in MySQL and checks whether stock has fallen below the predefined minimum level. When a low-stock condition is detected, the backend sends relevant information to the Gemini API to generate a supplier message. The proposed message is returned to the dashboard for human review and approval. Once approved, the configured sending process is initiated, and the action is recorded in the Activity_Logs table.

SDC Sponsor-Provided Tools: The project plans to use the Gemini API and Replit as part of its AI and cloud-hosting architecture. However, whether these services are officially provided or sponsored by SDC must be confirmed with the organisers. The team will declare the actual services and access arrangements used in the final implementation.


Fronted Dashboard
JavaScript/ Inventory/Approval
Backend Server
Node.js/API/ Stock Rules
MySQL
Product/ Suppliers/ Activity Logs
Gemini API
Generate supplier message
Human Approval > Send Message > Activity Log
Deployment: Replit

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team adopts a user-centred design-thinking approach to develop the Orbyte Smart Reorder Assistant for small grocery shops (*kedai runcit*) in Malaysia.

Empathise: We identify the challenges faced by grocery shop owners, including repetitive stock checking, manual inventory updates, and contacting multiple suppliers through WhatsApp.

Define: We focus on one specific problem: the time and effort required to monitor low-stock products and prepare replenishment messages.

Ideate: We propose an automated workflow that updates inventory, detects low-stock items, and uses the Gemini API to draft contextual supplier messages while keeping the shop owner in control.

Prototype: We plan to build a web-based dashboard using JavaScript, with a Node.js backend, MySQL database, and Replit hosting. The interface will display stock levels, low-stock alerts, message previews, and approval controls.

Test: We will test inventory updates, stock threshold detection, AI-generated messages, approval and rejection actions, and activity log records using synthetic data. Identified issues will be corrected before deployment.

For Phase 2, we aim to improve reliability, usability, and scalability through iterative testing. We will also review data protection practices, secure API credentials, and ensure that supplier messages require human approval before sending. These steps will help prepare the prototype for potential use by different small grocery businesses.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8 – Decent Work and Economic Growth** - **Target 8.2:** Achieve higher levels of economic productivity through technological upgrading and innovation. - **Target 8.3:** Promote productive activities, entrepreneurship, and the growth of micro, small, and medium-sized enterprises. The project supports these targets by automating repetitive inventory monitoring and supplier communication tasks for small grocery shops.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Estimated baseline: 60 minutes per operating day per shop owner** spent checking shelves, recording low-stock products, and preparing supplier messages manually. This is an illustrative estimate based on the Pak Abu scenario and must be validated through prototype testing or interviews with actual *kedai runcit* owners.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary KPI: Percentage reduction in time spent on inventory monitoring and supplier message preparation (%).** Measure the time required to complete the same inventory and replenishment tasks before and after using the prototype. Formula:

Metric Category	Your Response
	Time Reduction (%) = [(Baseline Time − Automated Time) / Baseline Time] × 100%
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Our preliminary target is to reduce the time spent on inventory monitoring and supplier message preparation by at least 30%. Based on an estimated baseline of 60 minutes per day, this would reduce the time required to approximately 42 minutes, saving 18 minutes daily. The 30% improvement is a proposed target rather than a verified result. We will validate its feasibility by comparing manual and automated task completion times using the same inventory scenarios during prototype testing. The target may be adjusted based on the results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js	Code / Runtime	MIT License
MySQL	Database Software	GNU GPL (Community Edition)
Google Gemini API	AI Model / API	Google's applicable API terms
Replit	Cloud Platform	Proprietary service
Aiven	Managed Database Service	Aiven service terms, database software license depends on the edition used.
WhatsApp Click-to-Chat	Messaging / Hosted Service	Free public feature; no account or API key

React	Code/framework	MIT license

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 25/10/2026